

LABORATORY 08 – Data Link Layer and Application Layer

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COMPUTER NETWORKS

GROUP 01

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OBJETIVE

Revisar y configurar la infraestructura LAN de una empresa, trabajando con redes Ethernet y WiFi, dispositivos de interconexión, y servicios de la capa de aplicación como servidores web, DNS, correo electrónico y bases de datos.

MATERIALS & TOOLS

- Laboratory computers with internet access.
- Virtualization software.
- Switches
- Wireshark & Packet tracer
- Wires type cat

INTRODUCTION

In this lab, we will continue developing a technological infrastructure, integrating wired and wireless workstations, physical and virtual servers, and network devices such as switches and routers. We will use tools such as Wireshark and Cisco Packet Tracer to analyze the behavior of Ethernet and Wi-Fi networks, as well as configure essential services at the application layer. We will also explore advanced configurations such as VLANs, secure wireless networks, and the implementation of dynamic web services with databases, thus strengthening our practical understanding of the data link and application layers.

THEORETICAL FRAMEWORK

Network hardware management is essential for the proper operation of any communications infrastructure. Devices such as switches, routers, wireless access points, and physical or virtual servers form the foundation upon which local area networks and their connection to external networks are built. Layer 2 and Layer 3 switches allow multiple devices to be interconnected in a network, facilitating segmentation and logical addressing of data. Routers, for their part, enable communication between different networks using routing protocols and IP address assignment.

In this context, understanding the basic configuration of these devices is crucial. This includes assigning names, setting access passwords, configuring interfaces, defining VLANs to segment the network, and applying security mechanisms such as access control lists and authentication methods in Wi-Fi networks such as WPA2-PSK.

Network analysis, on the other hand, allows for diagnosing network behavior and ensuring its proper functioning. Tools like Wireshark allow you to capture and examine network packets in real time, displaying details such as MAC addresses, protocols used, frame content, and possible transmission errors. This type of analysis is useful for verifying connectivity, understanding the operation of protocols such as ARP, DHCP, and TCP/IP, and detecting potential failures or bottlenecks.

Furthermore, the use of simulation environments such as Cisco Packet Tracer provides a safe space to design, test, and visualize network operations without the need for physical hardware. With these tools, it is possible to observe how devices build their routing or switching tables, how the spanning tree algorithm works to avoid loops, and how packets behave in networks with VLANs or wireless connections.

Together, mastering network hardware and analysis tools strengthens professionals' ability to build efficient, secure networks tailored to an organization's needs.

SETUP

In this section, the lab's wired and wireless networks were configured, including switches, PCs, and routers. Connectivity was verified, Ethernet frames were analyzed with Wireshark, VLANs were implemented to segment the network, and secure Wi-Fi networks were configured.

BASIC SWITCH CONFIGURATION

Just like in the previous lab, we configured the switch assigned to us by the instructor during the lab. To do this, we accessed the PuTTY console and entered the necessary commands to connect the ports on our table and its computers to the designated IP ranges.

```
Switch1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch1(config)#interface fastEthernet0/10
Switch1(config-if)#no shutdown
Switch1(config-if)#no ip domain lookup
Switch1(config)#interface fastEthernet0/3
Switch1(config-if)#no shutdown
Switch1(config-if)#no ip domain lookup
Switch1(config)#exit
Switch1#
```

```
nged state to up
Switch#ena
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Switch1
Switch1(config)#end
Switch1#write
*Mar  1 03:25:46.470: %SYS-5-CONFIG_I: Configured from console by co
Switch1#write memory
Building configuration...
[OK]
Switch1#
```

When configuring the system, we worked with a couple of colleagues to connect the switches via the rack and test connections with the `ping` command, thus validating that we have a small network between the four devices, as shown in the diagram.

<pre>C:\Users\Redes>ping 183.24.30.105 Pinging 183.24.30.105 with 32 bytes of data: Reply from 183.24.30.105: bytes=32 time=2ms TTL=128 Reply from 183.24.30.105: bytes=32 time=1ms TTL=128 Reply from 183.24.30.105: bytes=32 time=2ms TTL=128 Reply from 183.24.30.105: bytes=32 time=1ms TTL=128 Ping statistics for 183.24.30.105: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms</pre>	<pre>C:\Users\Redes>ping 183.24.30.106 Pinging 183.24.30.106 with 32 bytes of data: Reply from 183.24.30.106: bytes=32 time=1ms TTL=128 Reply from 183.24.30.106: bytes=32 time=2ms TTL=128 Reply from 183.24.30.106: bytes=32 time=2ms TTL=128 Reply from 183.24.30.106: bytes=32 time=2ms TTL=128 Ping statistics for 183.24.30.106: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms</pre>
---	---

At this point, we have a multi-device system that allows us to use traffic monitoring tools like Wireshark. In this part, we need to view the content of the intercepted frames and obtain MAC addresses, error control, and other information.

The image shows a Wireshark capture of network traffic. The packet list on the left shows several ICMP Echo (ping) requests and replies between 183.24.30.105 and 183.24.30.106, as well as ARP requests. The packet details pane on the right shows the structure of a selected ICMP Echo request, including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol fields. The packet bytes pane on the right shows the raw hex and ASCII data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco_5a:59:01	Spanning-tree-for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x8001
2	2.004460	Cisco_5a:59:01	Spanning-tree-for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x8001
3	2.035554	183.24.30.105	183.24.30.106	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 4)
4	2.037329	183.24.30.106	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=128 (request in 3)
5	3.050335	183.24.30.105	183.24.30.106	ICMP	74	Echo (ping) request id=0x0001, seq=35/8960, ttl=128 (reply in 6)
6	3.051872	183.24.30.106	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=35/8960, ttl=128 (request in 5)
7	4.013782	Cisco_5a:59:01	Spanning-tree-for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x8001
8	4.066148	183.24.30.105	183.24.30.106	ICMP	74	Echo (ping) request id=0x0001, seq=36/9216, ttl=128 (reply in 9)
9	4.067952	183.24.30.106	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=36/9216, ttl=128 (request in 8)
10	5.081760	183.24.30.105	183.24.30.106	ICMP	74	Echo (ping) request id=0x0001, seq=37/9472, ttl=128 (reply in 11)
11	5.083411	183.24.30.106	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=37/9472, ttl=128 (request in 10)
12	6.014602	Cisco_5a:59:01	Spanning-tree-for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x8001
13	6.001216	HewlettP_25:77:e4	HewlettP_25:77:e4	ARP	60	Who has 183.24.30.105? Tell 183.24.30.106
14	6.001252	HewlettP_25:77:e4	HewlettP_25:77:e4	ARP	42	183.24.30.105 is at 50:65:f3:25:77:e4
15	6.069320	HewlettP_25:77:e4	HewlettP_25:77:e4	ARP	42	Who has 183.24.30.106? Tell 183.24.30.105
16	6.070200	HewlettP_25:77:e4	HewlettP_25:77:e4	ARP	60	183.24.30.106 is at 50:65:f3:25:77:e4

Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{81EF03E3-78B1-4550-98FC-36A961870439}

Section number: 1

> Interface id: 0 (\Device\NPF_{81EF03E3-78B1-4550-98FC-36A961870439})

Encapsulation type: Ethernet (1)

Arrival Time: May 9, 2023 13:49:19.214099000 SA Pacific Standard Time

[Time shift for this packet: 0.000000000 seconds]

Epoch Time: 1683658159.214099000 seconds

[Time delta from previous captured frame: 0.031094000 seconds]

[Time delta from previous displayed frame: 0.031094000 seconds]

[Time since reference or first frame: 2.035554000 seconds]

Frame Number: 3

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: ethertypes:ip:icmp:data]

[Coloring Rule Name: ICMP]

[Coloring Rule String: icmp || icmpv6]

> Ethernet II, Src: HewlettP_25:77:e4 (50:65:f3:25:77:e4), Dst: HewlettP_25:77:e4 (50:65:f3:25:77:e4)

> Internet Protocol Version 4, Src: 183.24.30.105, Dst: 183.24.30.106

> Internet Control Message Protocol

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco_Sa:59:01	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x0001
2	2.004460	Cisco_Sa:59:01	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x0001
3	2.035554	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 4)
4	2.037329	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=128 (request in 3)
5	3.050335	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) request id=0x0001, seq=35/8960, ttl=128 (reply in 6)
6	3.051072	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=35/8960, ttl=128 (request in 5)
7	4.013782	Cisco_Sa:59:01	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x0001
8	4.066148	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) request id=0x0001, seq=36/9216, ttl=128 (reply in 9)
9	4.067952	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=36/9216, ttl=128 (request in 8)
10	5.061760	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) request id=0x0001, seq=37/9472, ttl=128 (reply in 11)
11	5.063411	183.24.30.105	183.24.30.105	ICMP	74	Echo (ping) reply id=0x0001, seq=37/9472, ttl=128 (request in 10)
12	6.014002	Cisco_Sa:59:01	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/1/00:24:51:5a:59:00 Cost = 0 Port = 0x0001
13	6.601216	HewlettP_25:7a:be	HewlettP_25:77:e4	ARP	60	Who has 183.24.30.105? Tell 183.24.30.105
14	6.601252	HewlettP_25:7a:be	HewlettP_25:77:e4	ARP	42	183.24.30.105 is at 50:65:f3:25:77:e4
15	6.669320	HewlettP_25:77:e4	HewlettP_25:7a:be	ARP	42	Who has 183.24.30.105? Tell 183.24.30.105
16	6.670000	HewlettP_25:7a:be	HewlettP_25:77:e4	ARP	60	183.24.30.105 is at 50:65:f3:25:77:e4

> Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{81EFD3E... > Ethernet II, Src: HewlettP_25:77:e4 (50:65:f3:25:77:e4), Dst: HewlettP_25:7a:be (50:65:f3:25:7a:be) > Destination: HewlettP_25:7a:be (50:65:f3:25:7a:be) > Source: HewlettP_25:77:e4 (50:65:f3:25:77:e4) > Type: IPv4 (0x0000) > Internet Protocol Version 4, Src: 183.24.30.105, Dst: 183.24.30.106 > Internet Control Message Protocol	<pre> 0000 50 65 f3 25 7a be 50 65 f3 25 77 e4 00 00 45 00 Pe-Kz-Pe-%m...L 0010 00 3c be f8 00 00 01 00 00 b7 18 1e 69 b7 18 --<-----1-- 0020 1e 6a 06 00 4d 39 00 01 00 22 61 62 63 64 65 66 -j-M9...-abcdef 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklm opqrstv 0040 77 61 62 63 64 65 66 67 68 69 wxyzdefg hi </pre>
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Finally, we just need to validate the connections with the other groups, since we already understand the internal structure of the frames. Therefore, we use ping between the four connected computers and validate it's functionality.

```

C:\Users\Redes>ping 183.24.30.107

Pinging 183.24.30.107 with 32 bytes of data:
Reply from 183.24.30.107: bytes=32 time=3ms TTL=128
Reply from 183.24.30.107: bytes=32 time=2ms TTL=128
Reply from 183.24.30.107: bytes=32 time=2ms TTL=128
Reply from 183.24.30.107: bytes=32 time=2ms TTL=128

Ping statistics for 183.24.30.107:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\Redes>ping 183.24.30.108

Pinging 183.24.30.108 with 32 bytes of data:
Reply from 183.24.30.108: bytes=32 time=3ms TTL=128
Reply from 183.24.30.108: bytes=32 time=2ms TTL=128
Reply from 183.24.30.108: bytes=32 time=2ms TTL=128
Reply from 183.24.30.108: bytes=32 time=2ms TTL=128

Ping statistics for 183.24.30.108:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

```

```

C:\Users\Redes>ping 183.24.30.107

Pinging 183.24.30.107 with 32 bytes of data:
Reply from 183.24.30.107: bytes=32 time=1ms TTL=128
Reply from 183.24.30.107: bytes=32 time=2ms TTL=128
Reply from 183.24.30.107: bytes=32 time=2ms TTL=128
Reply from 183.24.30.107: bytes=32 time=1ms TTL=128

Ping statistics for 183.24.30.107:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Redes>ping 183.24.30.108

Pinging 183.24.30.108 with 32 bytes of data:
Reply from 183.24.30.108: bytes=32 time=1ms TTL=128
Reply from 183.24.30.108: bytes=32 time=1ms TTL=128
Reply from 183.24.30.108: bytes=32 time=2ms TTL=128
Reply from 183.24.30.108: bytes=32 time=2ms TTL=128

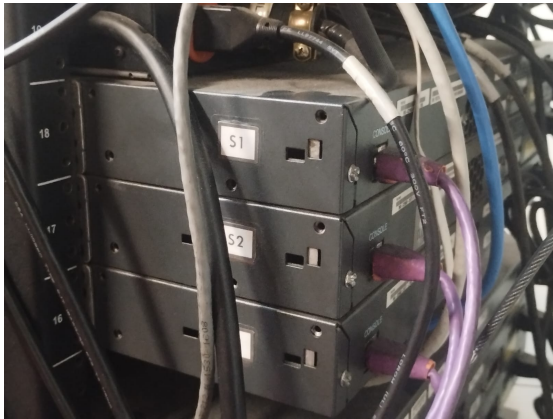
Ping statistics for 183.24.30.108:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Redes>

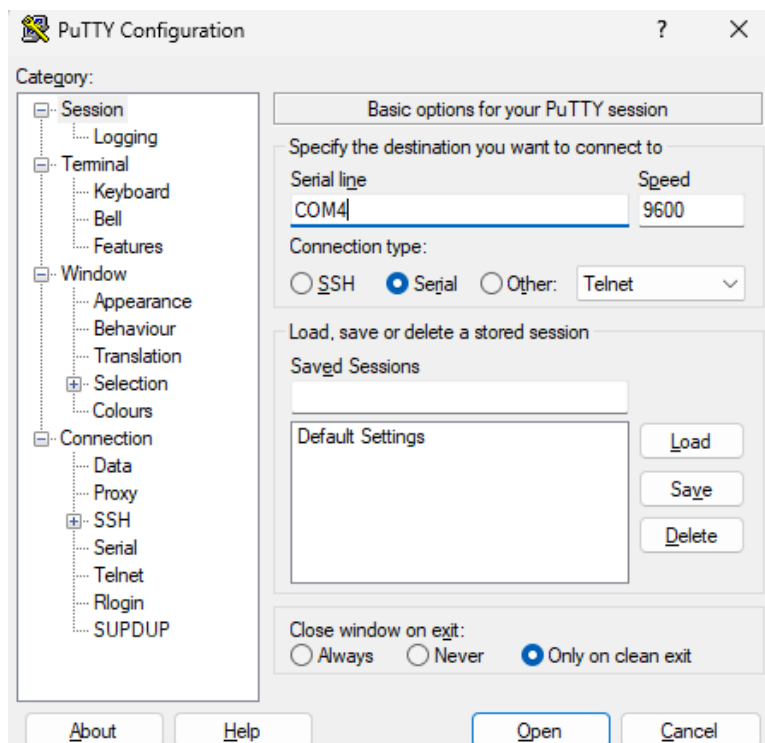
```

BASIC SWITCH CONFIGURATION II

For this section, and as in the previous lab, we'll configure the routers to begin testing the interconnections based on the lab switches. To do this, we'll repeat the same configuration procedure, starting with logging into PuTTY. Obviously, we need to connect the console cables to our table in the rack.



This way, we can start the application we're connecting to using the SSH protocol. In this case, we simply select the serial connection and the connector name, and it should work.



Now we need to execute the commands required to complete the configurations requested in the lab guide. These allow us to connect to other routers through the already configured switches.

```
Router>
Router>enable
Router#configure terminal Espinosa
      ^
% Invalid input detected at '^' marker.

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Espinosa
Espinosa(config)#enable
% Incomplete command.

Espinosa(config)#clear
      ^
% Invalid input detected at '^' marker.

Espinosa(config)#show version
      ^
% Invalid input detected at '^' marker.

Espinosa(config)#enable password cisco
Espinosa(config)#line console 0
Espinosa(config-line)#password claveC
Espinosa(config-line)#login
Espinosa(config-line)#logging synchronouns
Translating "synchronouns"...domain server (255.255.255.255)
      ^
% Invalid input detected at '^' marker.

Espinosa(config-line)#logging synchronous
Espinosa(config-line)#exit
Espinosa(config)#line vty 0 4
Espinosa(config-line)#password claveT
Espinosa(config-line)#login
Espinosa(config-line)#logging synchornous
Translating "synchornous"...domain server (255.255.255.255)
      ^
% Invalid input detected at '^' marker.

Espinosa(config-line)#logging synchronous
Espinosa(config-line)#exit
Espinosa(config)#no ip domain-lookup
Espinosa(config)#banner motd #Buenas, practica lab 07#
Espinosa(config)#
```



```

Espinosa(config)#show version
^
% Invalid input detected at '^' marker.

Espinosa(config)#enable password cisco
Espinosa(config)#line console 0
Espinosa(config-line)#password claveC
Espinosa(config-line)#login
Espinosa(config-line)#logging synchronouns
Translating "synchronouns"...domain server (255.255.255.255)
^
% Invalid input detected at '^' marker.

Espinosa(config-line)#logging synchronous
Espinosa(config-line)#exit
Espinosa(config)#line vty 0 4
Espinosa(config-line)#password claveT
Espinosa(config-line)#login
Espinosa(config-line)#logging synchornous
Translating "synchornous"...domain server (255.255.255.255)
^
% Invalid input detected at '^' marker.

Espinosa(config-line)#logging synchronous
Espinosa(config-line)#exit
Espinosa(config)#no ip domain-lookup
Espinosa(config)#banner motd #Buenas, practica lab 07#
Espinosa(config)#interface GigabitB?
% Unrecognized command

```

```

Espinosa(config)#interface GigabitEthernet0/0
Espinosa(config-if)#description #Equipo de 600 hosts
Espinosa(config-if)#ip address 88.0.4.0 255.255.252.0
Bad mask /22 for address 88.0.4.0
Espinosa(config-if)#ip address 88.0.4.1 255.255.252.0
Espinosa(config-if)#interface GigabitEthernet0/1
Espinosa(config-if)#description #Equipo de 4000 hosts
Espinosa(config-if)#ip address 88.0.16.1 255.255.240.0
Espinosa(config-if)#no shutdown
Espinosa(config-if)#exit
Espinosa(config)#description #Equipo de 4000 hosts
*Apr  7 22:26:53.911: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
Espinosa(config)#
*Apr  7 22:27:02.151: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Apr  7 22:27:03.151: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Espinosa(config)#interface GigabitEthernet0/0
^
% Invalid input detected at '^' marker.

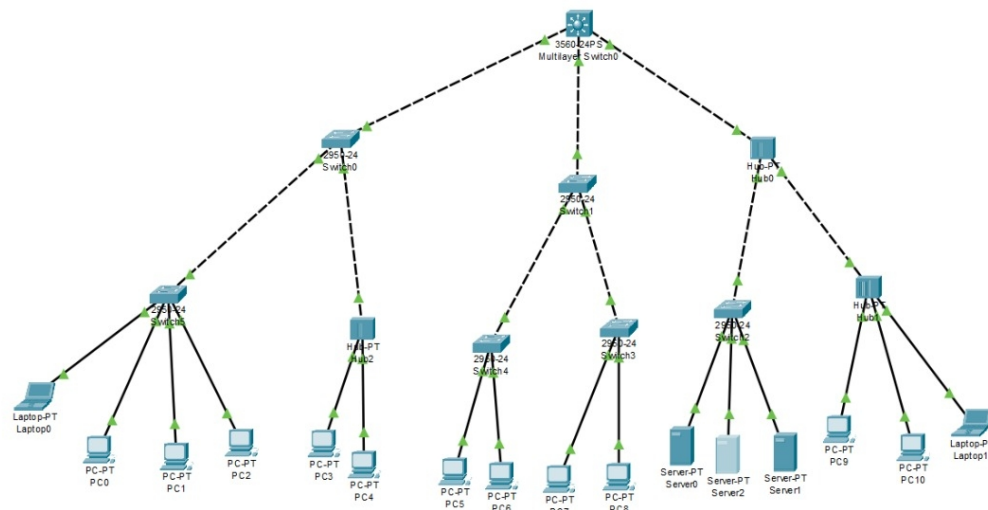
Espinosa(config)#interface GigabitEthernet0/0
Espinosa(config-if)#description #Equipo de 600 hosts
Espinosa(config-if)#ip address 88.0.4.1 255.255.252.0
Espinosa(config-if)#no shutdown
Espinosa(config-if)#exit
Espinosa(config)#
*Apr  7 22:27:33.487: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
Espinosa(config)#
*Apr  7 22:27:42.151: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Apr  7 22:27:43.151: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Espinosa(config)#

```

This section doesn't explain much because it's something we'd already covered in the previous lab, as mentioned above. Even so, we decided to include a quick explanation for general understanding of the router and switches configuration.

LARGER SWITCH NETWORKS

In this first section of Packet Tracer, we'll understand how switches and some related algorithms work. To do this, we need to replicate the diagram shown in the guide.



Now, we need to perform the configurations we saw in previous labs on all the switches. Additionally, we configure the devices based on the ranges and parameters in the guide, so that each device accesses the same network.

```
Switch0#
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
banner motd ^C Uso exclusivo para estudantes de RECO Lab9 ^C
!
!
!
line con 0
password Clave_C
logging synchronous
login
!
line vty 0 4
password Clave_T
logging synchronous
login
line vty 5 15
password Clave_T
logging synchronous
login
!
!
end
```

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

124.48.87.40

Subnet Mask

255.255.255.0

Default Gateway

124.48.87.1

DNS Server

0.0.0.0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

124.48.87.10

Subnet Mask

255.255.255.0

Default Gateway

124.48.87.1

DNS Server

0.0.0.0

We validate the equipment connection by pinging between the first and seventh equipment in our Packet Tracer network.

Laptop0

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 124.48.87.55

Pinging 124.48.87.55 with 32 bytes of data:

Reply from 124.48.87.55: bytes=32 time=11ms TTL=128
Reply from 124.48.87.55: bytes=32 time=10ms TTL=128
Reply from 124.48.87.55: bytes=32 time=126ms TTL=128
Reply from 124.48.87.55: bytes=32 time=10ms TTL=128

Ping statistics for 124.48.87.55:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 126ms, Average = 39ms

C:\>ping 124.48.87.47

Pinging 124.48.87.47 with 32 bytes of data:

Reply from 124.48.87.47: bytes=32 time<1ms TTL=128
Reply from 124.48.87.47: bytes=32 time<1ms TTL=128
Reply from 124.48.87.47: bytes=32 time<1ms TTL=128
Reply from 124.48.87.47: bytes=32 time=10ms TTL=128

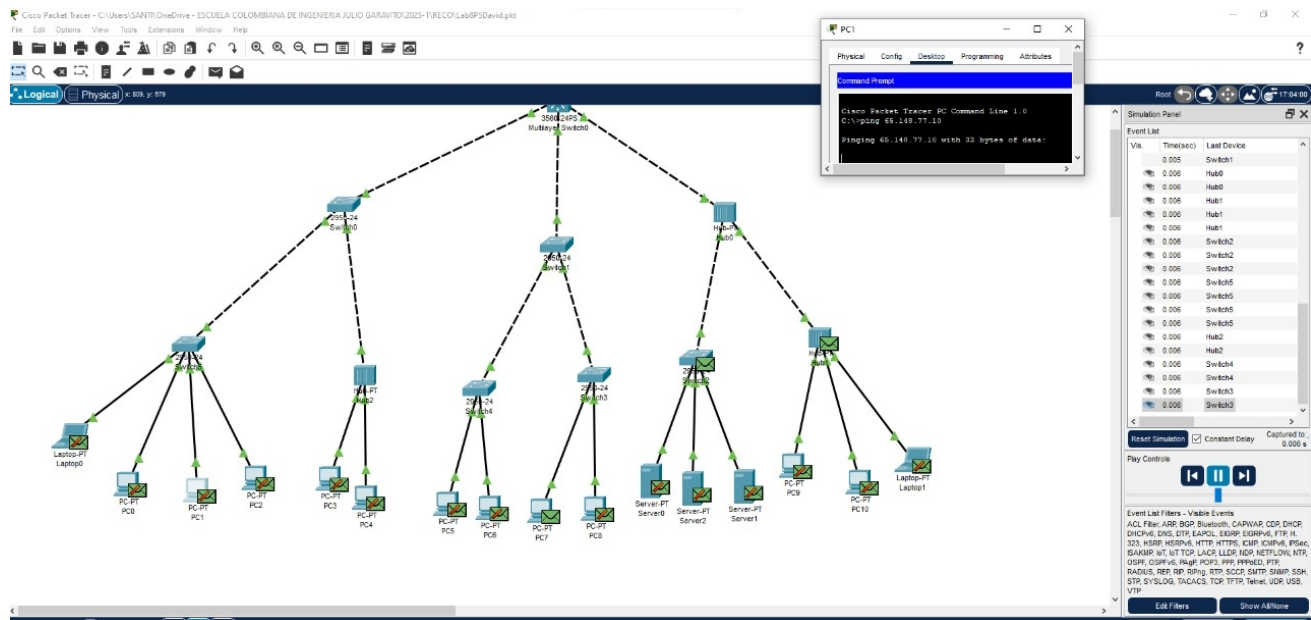
Ping statistics for 124.48.87.47:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>

☐ Top

At this point, using simulation mode, we review the network behavior and the format of an Ethernet frame when sending frames. We identify the behavior of the switches and the hash tables for each of the requested cases:

- From PC1 to PC7:



Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```

Press RETURN to get started!

Use exclusivo para estudiantes de RECO Lab9

User Access Verification

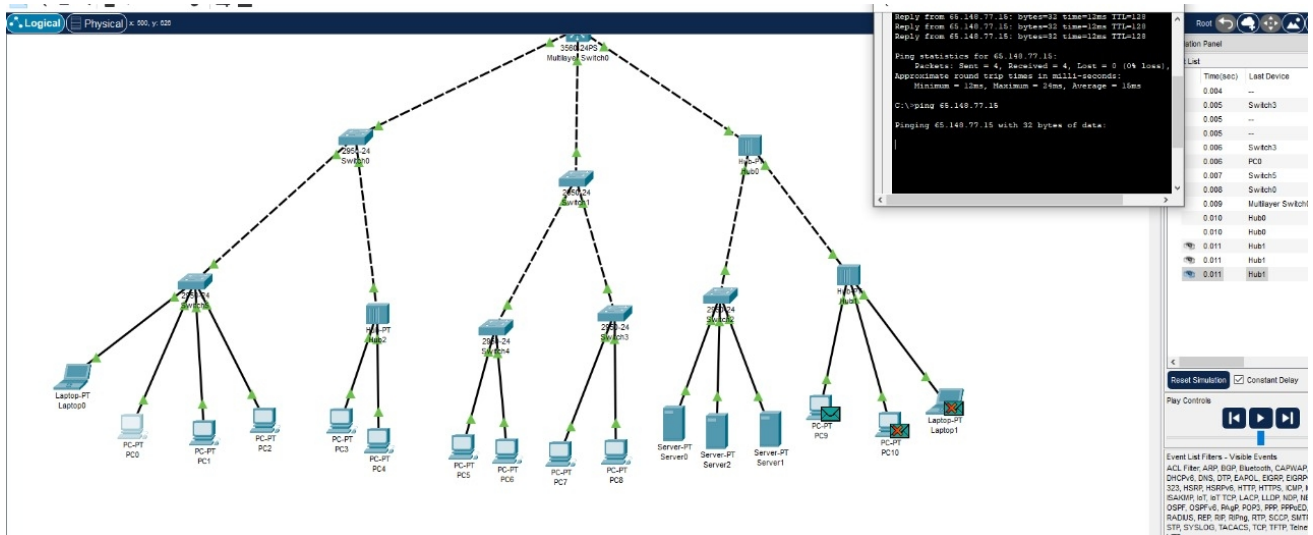
Password:

Switch3>enable
Password:
Switch3#show mac add
Switch3#show mac address-table
        Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0030.f20e.6874   DYNAMIC Gi0/1
1       0060.47b7.c219   DYNAMIC Gi0/1
1       00d0.d3b6.a49a   DYNAMIC Fa0/3
Switch3#
  
```

Copy Paste

☐ Top

- From PC0 to PC9:



Switch3
Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

Uso exclusivo para estudiantes de RECO Lab9

User Access Verification

Password:

Switch3>enable

Password:

Switch3#show mac add

Switch3#show mac address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
1	0030.f20e.6874	DYNAMIC	Gig0/1
1	0060.47b7.c219	DYNAMIC	Gig0/1
1	00d0.d3b6.a49a	DYNAMIC	Fa0/3

Switch3#show mac address-table

Mac Address Table

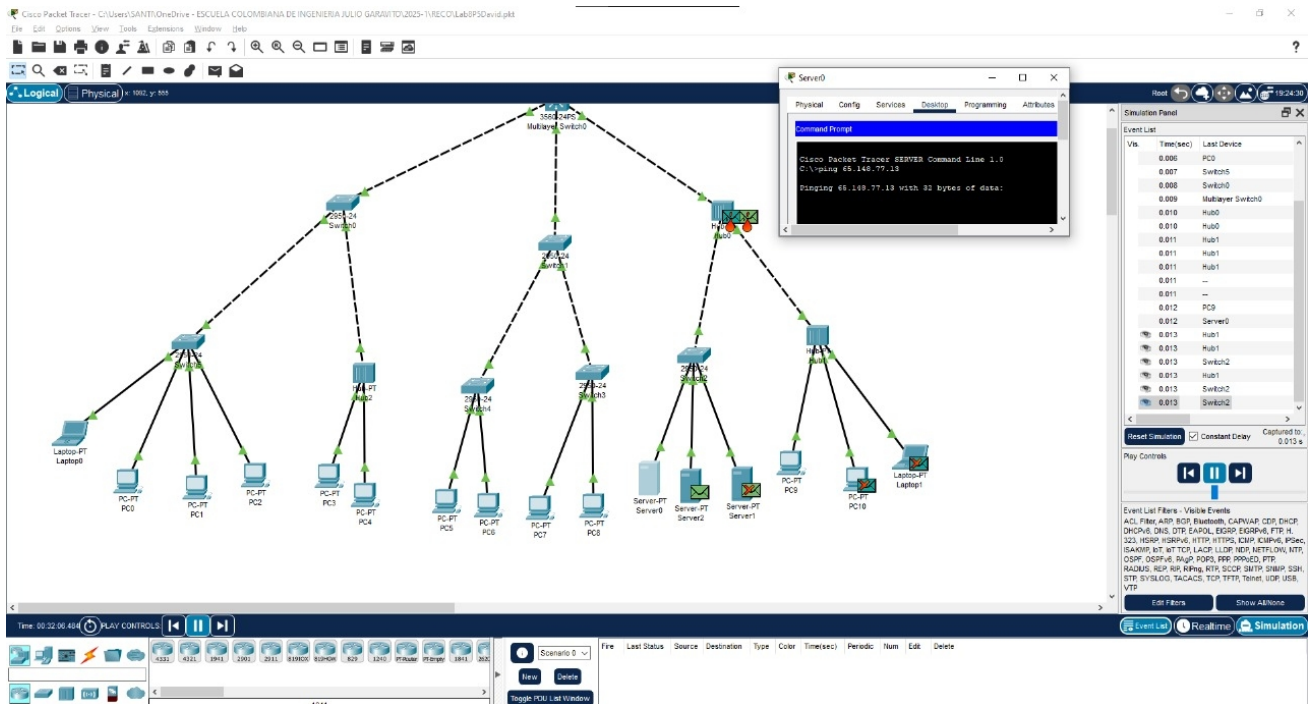
Vlan	Mac Address	Type	Ports
1	000a.f3b1.a45c	DYNAMIC	Gig0/1
1	0030.f20e.6874	DYNAMIC	Gig0/1
1	0060.47b7.c219	DYNAMIC	Gig0/1
1	0090.2b9c.7b33	DYNAMIC	Fa0/2
1	00d0.d3b6.a49a	DYNAMIC	Fa0/3

Switch3#

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☐ Top

- From Server0 to Server1:



Switch5

Physical Config CLI Attributes

IOS Command Line Interface

```

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User Access Verification

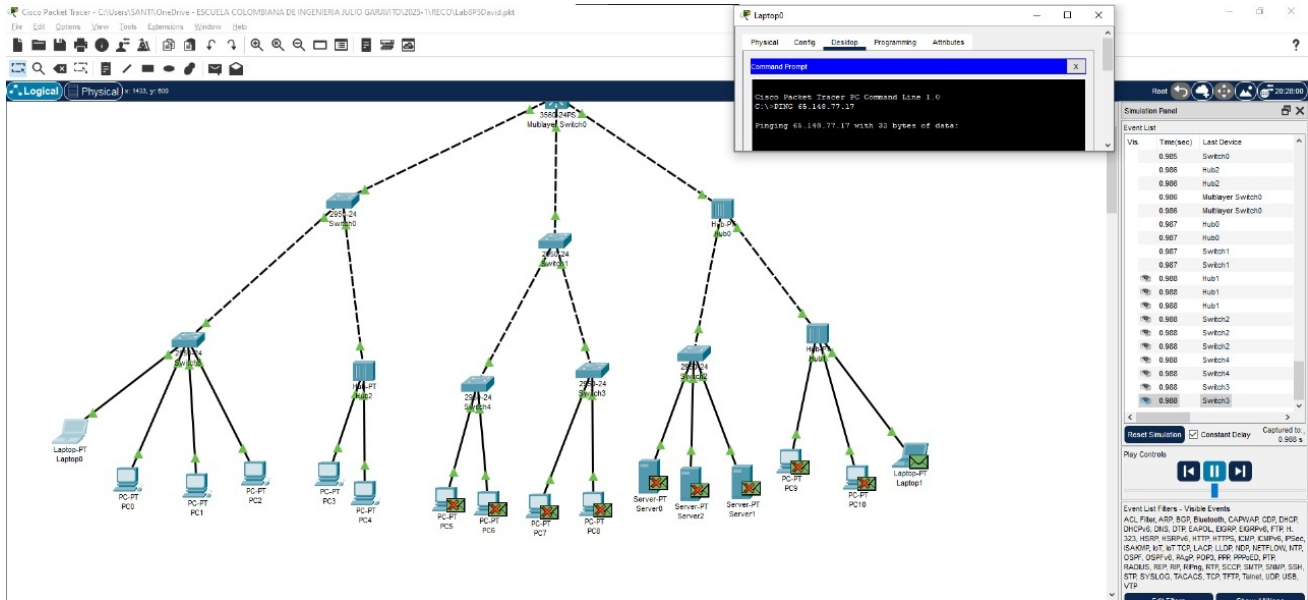
Password:

Switch5>enable
Password:
Switch5#show mac ta
Switch5#show mac add
Switch5#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0001.4337.a2ae   DYNAMIC   Fa0/2
1       000a.f3b1.a45c   DYNAMIC   Gig0/1
1       0060.2f2c.e612   DYNAMIC   Fa0/1
1       0090.2b5c.7b33   DYNAMIC   Gig0/1
1       00d0.baab.2e03   DYNAMIC   Gig0/1
1       00d0.d3b6.a49a   DYNAMIC   Gig0/1
Switch5#
  
```

Copy Paste

☐ Top

- From Laptop0 to Laptop1:



Switch3
— □ ×

Physical
Config
CLI
Attributes

IOS Command Line Interface

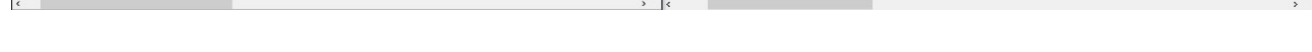
```

Vlan      Mac Address      Type      Ports
----      -
1         0030.f20e.6874   DYNAMIC   Gig0/1
1         0060.47b7.c219   DYNAMIC   Gig0/1
1         00d0.d3b6.a49a   DYNAMIC   Fa0/3
Switch3#show mac address-table
      Mac Address Table
-----
Vlan      Mac Address      Type      Ports
----      -
1         000a.f3b1.a45c   DYNAMIC   Gig0/1
1         0030.f20e.6874   DYNAMIC   Gig0/1
1         0060.47b7.c219   DYNAMIC   Gig0/1
1         0090.2b9c.7b33   DYNAMIC   Fa0/2
1         00d0.d3b6.a49a   DYNAMIC   Fa0/3
Switch3#show mac address-table
      Mac Address Table
-----
Vlan      Mac Address      Type      Ports
----      -
1         0000.0c20.eac6   DYNAMIC   Fa0/1
1         000a.f3b1.a45c   DYNAMIC   Gig0/1
1         0030.f20e.6874   DYNAMIC   Gig0/1
1         0060.2f2c.e512   DYNAMIC   Gig0/1
1         0060.47b7.c219   DYNAMIC   Gig0/1
1         0090.2b9c.7b33   DYNAMIC   Fa0/2
1         00d0.97ac.2524   DYNAMIC   Gig0/1
1         00d0.d3b6.a49a   DYNAMIC   Fa0/3
Switch3#
          
```

Copy
Paste

☐ Top

C:\Users\SANTI\OneDrive - ESCUELA COLOMBIANA DE INGENIERIA\BOLIVAR\GABRIEL\2015-1\BOLIVAR\LABORATORIO\	1	Y	C:\Users\SANTI\OneDrive - ESCUELA COLOMBIANA DE INGENIERIA\BOLIVAR\GABRIEL\2015-1\BOLIVAR\LABORATORIO\	1	Y
--	---	---	--	---	---



VLAND CONFIGURATION

To this section and considering the reference diagram, we have to create VLANs so we can connect and communicate with different devices within the room. In this case, we have two VLANs with IDs 50 and 55, respectively.

Initially, we need to configure our router, as access to this network is done manually, using the `'access vland'` command on each network interface representing a device.

```
Switch1(config-if)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch1(config-if)#exit
Switch1(config)#interface fastEthernet0/2
Switch1(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Switch1(config-if)#exit
Switch1(config)#end
Switch1#
*Mar  1 04:41:13.634: %SYS-5-CONFIG_I: Configured from console by console
Switch1#write memory
Building configuration...
[OK]
Switch1#
```

We can view network information using the `'show vland brief'` command, where we validate which ones are active and which ports are assigned to them.

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gi0/1, Gi0/2
10	VLAN0010	active	Fa0/8
20	VLAN0020	active	Fa0/2
50	sistemas	active	
55	otros	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch1#
```

Now we configure the port to be a trunk, which is a point-to-point link between two network devices that carries more than one VLAN. This basically allows multiple ports to connect to a single point without overload the net.

```

interface GigabitEthernet0/1
  description "Conexion a switch 1"
  switchport mode trunk
!
interface GigabitEthernet0/2
  description "Conexion a switch 2"
  switchport mode trunk

```

Finally, we can perform a connectivity test by pinging the console to a machine that is on the same VLAN as each device, keeping in mind that the entire class has its respective devices on the corresponding network.

```

Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 1ms, Maximum = 1ms, Average = 1ms
Control-C
^C
C:\Users\Redes>ping 10.2.77.16

Pinging 10.2.77.16 with 32 bytes of data:
Reply from 10.2.77.16: bytes=32 time<1ms TTL=128
Reply from 10.2.77.16: bytes=32 time=1ms TTL=128

Ping statistics for 10.2.77.16:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
Control-C
^C
C:\Users\Redes>ping 10.2.77.17

Pinging 10.2.77.17 with 32 bytes of data:
Reply from 10.2.77.17: bytes=32 time=2ms TTL=128
Reply from 10.2.77.17: bytes=32 time=1ms TTL=128

Ping statistics for 10.2.77.17:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

```

```

Pinging 10.2.77.12 with 32 bytes of data:
Reply from 10.2.77.12: bytes=32 time=1ms TTL=128
Reply from 10.2.77.12: bytes=32 time=1ms TTL=128

Ping statistics for 10.2.77.12:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

```

```

C:\Users\Redes>ping 10.2.77.16

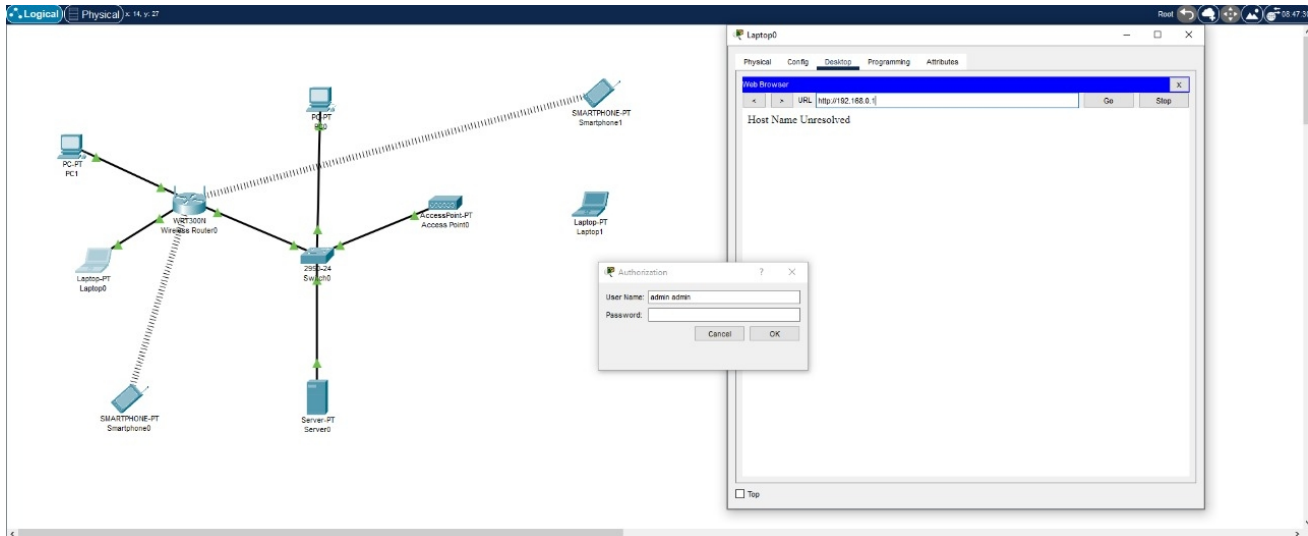
Pinging 10.2.77.16 with 32 bytes of data:
Reply from 10.2.77.16: bytes=32 time=1ms TTL=128
Reply from 10.2.77.16: bytes=32 time=1ms TTL=128
Reply from 10.2.77.16: bytes=32 time=1ms TTL=128
Reply from 10.2.77.16: bytes=32 time=1ms TTL=128

Ping statistics for 10.2.77.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

```

BASIC WIFI CONFIGURATION

In this section, we'll use Packet Tracer to represent and analyze wireless networks using the structure suggested in the guide. To do it, we need to follow the parameter indications for each of the elements in the diagram with a laptop device.



Using the laptop, we'll configure the requested settings by connecting to the router via the terminal. We'll configure the wired network to ensure a secure connection and also enter the subnet information to ensure proper operation. On the other hand, configure the wireless network with the same parameters so that our entire system is under the same configuration.

The screenshot shows the 'Web Browser' configuration page for a 'Wireless-N Broadband Router'. The URL bar shows 'http://192.168.0.1'. The page has tabs for 'Setup', 'Wireless', 'Security', 'Access Restrictions', and 'Applications & Gaming'. The 'Setup' tab is active, showing 'Internet Setup' and 'Network Setup' sections. Under 'Internet Setup', 'Internet Connection type' is set to 'Automatic Configuration - DHCP'. Under 'Network Setup', 'Router IP' is set to '192.168.0.1' with a 'Subnet Mask' of '255.255.255.0'. The 'DHCP Server Settings' section shows 'DHCP Server' as 'Enabled' and 'Start IP Address' as '192.168.0.100'.

Web Browser

URL: http://192.168.0.1

Go Stop

Internet Setup

Internet Connection type: Static IP

Internet IP Address: 192 . 168 . 0 . 0

Subnet Mask: 255 . 255 . 255 . 0

Default Gateway: 0 . 0 . 0 . 0

DNS 1: 0 . 0 . 0 . 0

DNS 2 (Optional): 0 . 0 . 0 . 0

DNS 3 (Optional): 0 . 0 . 0 . 0

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP

IP Address: 192 . 168 . 0 . 1

Subnet Mask: 255.255.255.192

Now we configure the wireless network repeater, this way we can validate connections with devices such as the SSID, network and router IP, range, WPA2-PSK authentication with AES, passwords with SECURITY_R and other settings.

Access Point0

Physical Config Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status: ☒ On

SSID: Espinosa_AP

2.4 GHz Channel: 6

Coverage Range (meters): 140,00

Authentication:

☐ Disabled ☐ WEP ☒ WPA2-PSK

WEP Key:

PSK Pass Phrase: SECURITY_AP

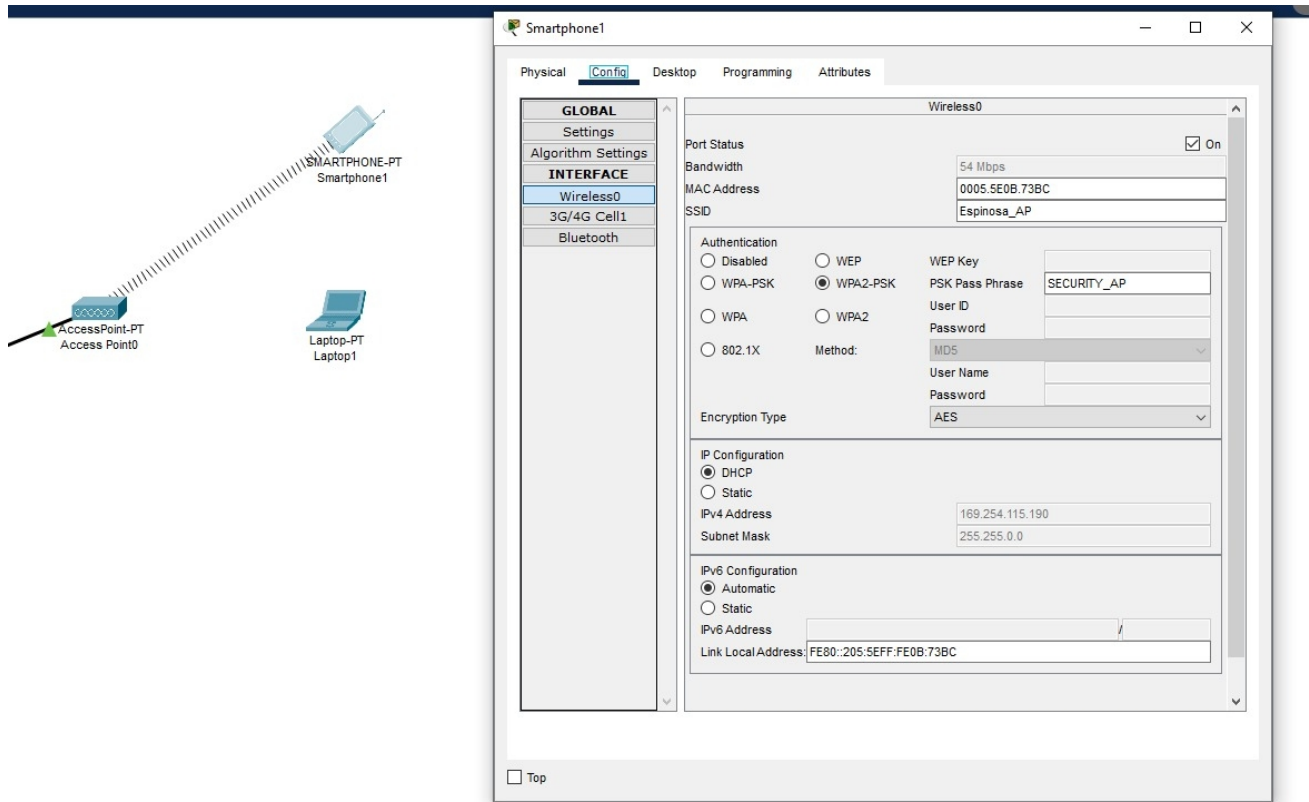
User ID:

Password:

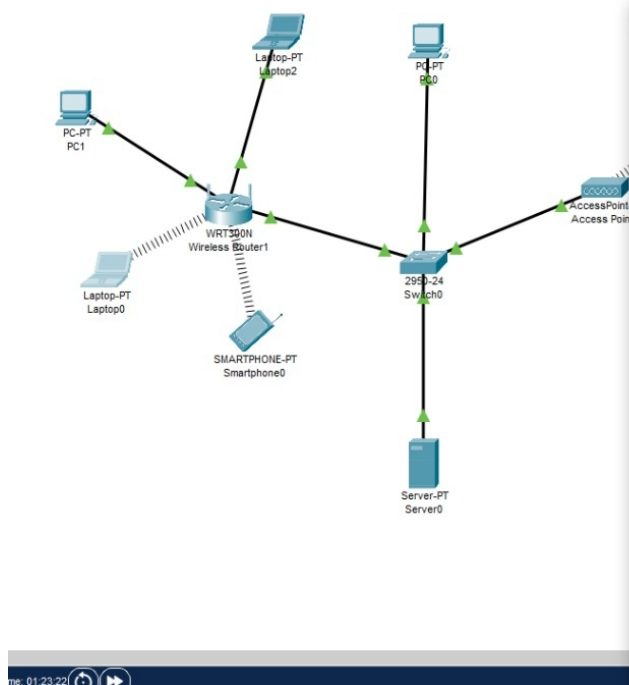
Encryption Type: AES

☐ Top

To finish the setup, we need to connect the wireless devices to the network we previously configured. We set the same parameters as for the repeater, but focusing on the network connection.



Finally, we ping between devices on the two present networks (wired and wireless) to validate the correct operation of the Packet Tracer setup,



```

Laptop0
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 65.148.77.254: bytes=32 time=26ms TTL=255
Reply from 65.148.77.254: bytes=32 time=15ms TTL=255

Ping statistics for 65.148.77.254:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 26ms, Average = 20ms

Control-C
~C
C:\>ping 65.148.77.2

Pinging 65.148.77.2 with 32 bytes of data:

Ping statistics for 65.148.77.2:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

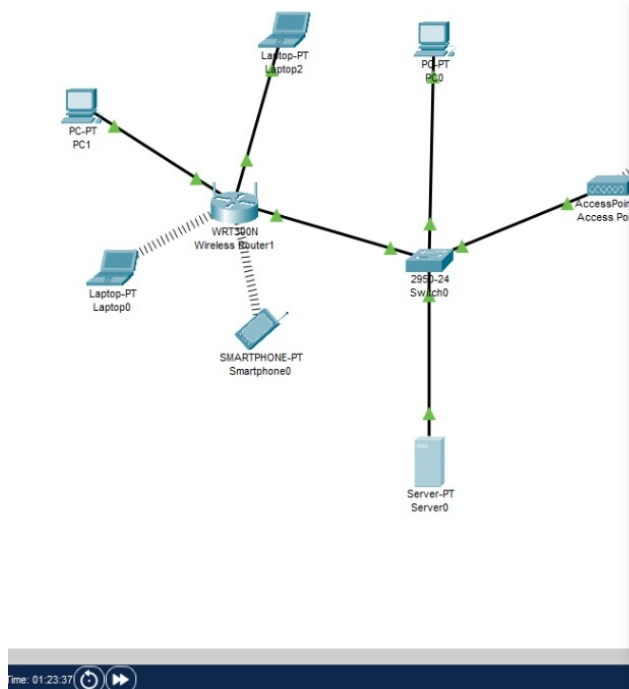
Control-C
~C
C:\>ping 192.168.0.100

Pinging 192.168.0.100 with 32 bytes of data:
Reply from 192.168.0.100: bytes=32 time=41ms TTL=128
Reply from 192.168.0.100: bytes=32 time=12ms TTL=128
Reply from 192.168.0.100: bytes=32 time=24ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 41ms, Average = 25ms

Control-C
~C
C:\>

```



```

Server0
Physical Config Services Desktop Programming Attributes
Command Prompt
Reply from 65.148.77.2: bytes=32 time<1ms TTL=128
Reply from 65.148.77.2: bytes=32 time<1ms TTL=128

Ping statistics for 65.148.77.2:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
~C
C:\>ping 65.148.77.2

Pinging 65.148.77.2 with 32 bytes of data:
Reply from 65.148.77.2: bytes=32 time<1ms TTL=128
Reply from 65.148.77.2: bytes=32 time<1ms TTL=128

Ping statistics for 65.148.77.2:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
~C
C:\>ping 65.148.77.100

Pinging 65.148.77.100 with 32 bytes of data:
Reply from 65.148.77.100: bytes=32 time=40ms TTL=128
Reply from 65.148.77.100: bytes=32 time=15ms TTL=128

Ping statistics for 65.148.77.100:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 40ms, Average = 27ms

Control-C
~C
C:\>

```

NI

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.100.80

Subnet Mask

255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address:

FE80::2D0:BCFF:FE47:41DD

- Green areas:

- Green areas:

Wireless0	
Port Status	☒ On
Bandwidth	36 Mbps
MAC Address	00D0.BC47.41DD
SSID	Rectangle
Authentication ☐ Disabled ☐ WEP ☐ WPA-PSK ☒ WPA2-PSK ☐ WPA ☐ WPA2 ☐ 802.1X Method: MD5 WEP Key PSK Pass Phrase recoreco User ID Password User Name Password	
Encryption Type	AES
IP Configuration ☐ DHCP ☒ Static IPv4 Address 192.168.100.80 Subnet Mask 255.255.255.0	
IPv6 Configuration ☐ Automatic ☒ Static IPv6 Address Link Local Address: FE80::2D0:BCFF:FE47:41DD	

- Purple areas:

Wireless0	
Port Status	☒ On
Bandwidth	54 Mbps
MAC Address	0003.E457.D261
SSID	Circle
Authentication ☐ Disabled ☐ WEP ☐ WPA-PSK ☒ WPA2-PSK ☐ WPA ☐ WPA2 ☐ 802.1X Method: MD5 WEP Key PSK Pass Phrase recoreco User ID Password User Name Password	
Encryption Type	AES
IP Configuration ☐ DHCP ☒ Static IPv4 Address 198.169.0.5 Subnet Mask 255.255.255.0	
IPv6 Configuration ☐ Automatic ☒ Static IPv6 Address Link Local Address: FE80::203:E4FF:FE57:D261	

This setup itself only allows you to connect devices that have a common switch, as seen in the image. However, for more remote or wireless networks, it doesn't allow connections without additional configuration, although it follows the same pattern of connecting common switches.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 171.18.100.50

Pinging 171.18.100.50 with 32 bytes of data:

Ping statistics for 171.18.100.50:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 171.18.100.51

Pinging 171.18.100.51 with 32 bytes of data:

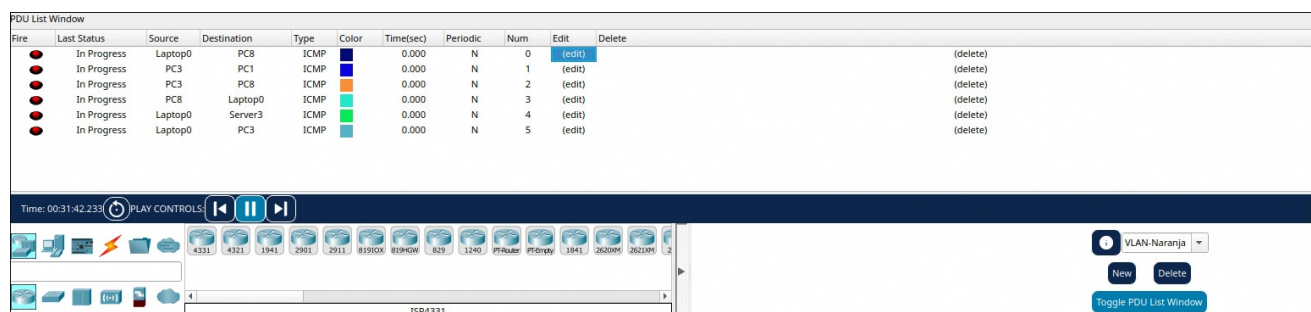
Reply from 171.18.100.51: bytes=32 time=1ms TTL=128
Reply from 171.18.100.51: bytes=32 time=11ms TTL=128
Reply from 171.18.100.51: bytes=32 time<1ms TTL=128

Ping statistics for 171.18.100.51:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 4ms

Control-C
^C
C:\>
```

Now that we have a functional setup, we need to configure the VLANs based on the colors determined in the guide and the created diagram. In this case, VLAN 50 was assigned to the green ones, VLAN 55 to the purple ones, and VLAN 60, resulting in the following virtual network zones.

- Naranja:



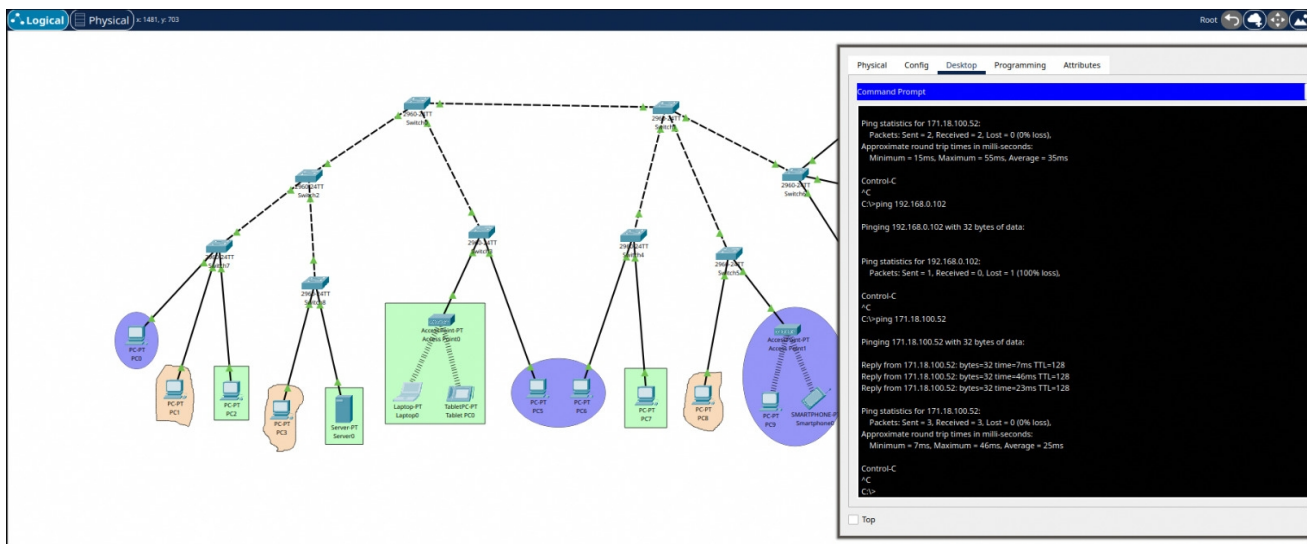
- Verde:

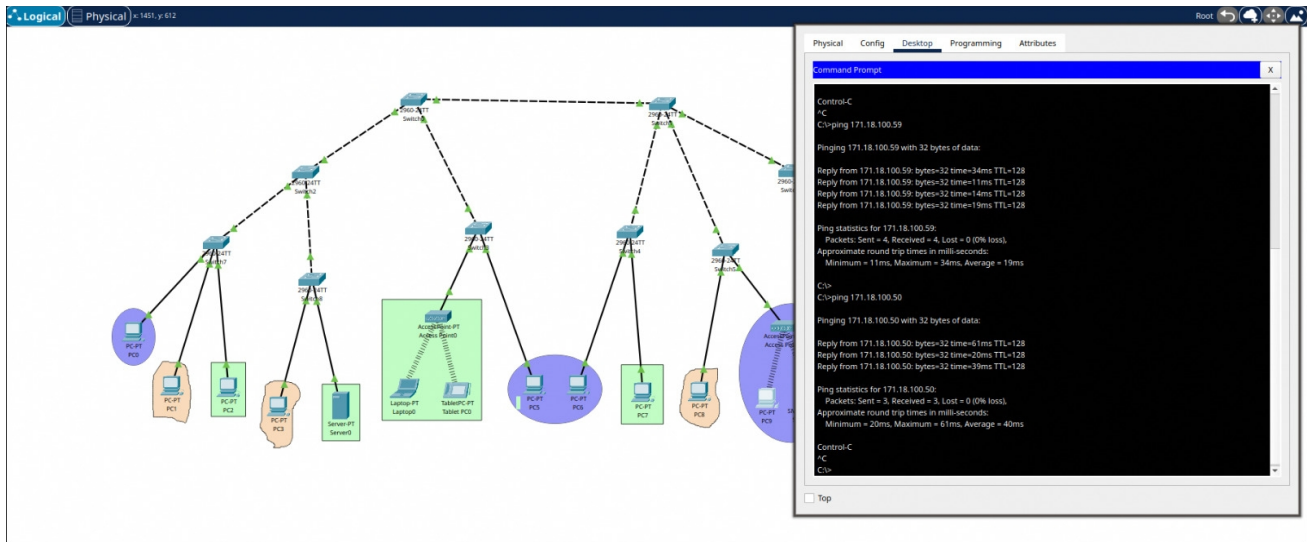
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	In Progress	Smartph...	PC7	ICMP		0.000	N	0	(edit)	(delete)
	In Progress	PC7	Server0	ICMP		0.000	N	1	(edit)	(delete)
	In Progress	Server0	PC2	ICMP		0.000	N	2	(edit)	(delete)
	In Progress	PC2	PC7	ICMP		0.000	N	3	(edit)	(delete)
	In Progress	Server2	Server0	ICMP		0.000	N	4	(edit)	(delete)
	In Progress	Tablet PC1	Tablet PC0	ICMP		0.000	N	5	(edit)	(delete)
	In Progress	Laptop1	Server1	ICMP		0.000	N	6	(edit)	(delete)
	In Progress	Server0	Server1	ICMP		0.000	N	7	(edit)	(delete)
	In Progress	PC2	Laptop0	ICMP		0.000	N	8	(edit)	(delete)

- Morado:

PDU List Window										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	In Progress	PC0	PC9	ICMP		0.000	N	0	(edit)	(delete)
	In Progress	PC0	Server1	ICMP		0.000	N	1	(edit)	(delete)
	In Progress	PC0	Smartphone0	ICMP		0.000	N	2	(edit)	(delete)
	In Progress	PC0	PC2	ICMP		0.000	N	3	(edit)	(delete)
	In Progress	PC0	PC5	ICMP		0.000	N	4	(edit)	(delete)
	In Progress	PC0	PC6	ICMP		0.000	N	5	(edit)	(delete)
	In Progress	PC0	PC7	ICMP		0.000	N	6	(edit)	(delete)
	In Progress	PC0	Server3	ICMP		0.000	N	7	(edit)	(delete)

Finally, we test by pinging between devices that are under the same VLAN, otherwise the connection is not recognized.





WIFI

In this section, we have a physical router for wireless devices. The trick is to connect it, configure it, and validate its operation using our mobile device configuration. Initially, we just need to connect the router and turn it on. Then, we connect one leg to the university network with crossed Ethernet cables and the other to the computer we'll be using to configure it.

When we access the URL (tplinklogin.net) provided by the router manufacturer, we'll be asked for credentials, which are the admin username and password.



Once inside, we'll see a general dashboard containing all the information about the router and the network to which it's connected. We'll also see different tabs to manipulate these settings.

TP-LINK®

300M Wireless N Router
Model No. TL-WR941ND

Status

Quick Setup

QSS

Network

Wireless

DHCP

Forwarding

Security

Parental Control

Access Control

Advanced Routing

Bandwidth Control

IP & MAC Binding

Dynamic DNS

System Tools

Status

Firmware Version: 3.13.9 Build 120201 Rel.54955n
Hardware Version: VR941N v2v3 0000000

LAN

MAC Address: 90-F6-52-69-7C-B2
IP Address: 192.168.0.1
Subnet Mask: 255.255.255.0

Wireless

Wireless Radio: Enable
Name (SSID): TP-LINK_897CB2
Channel: Auto (Current channel 4)
Mode: 11bgn mixed
Channel Width: Automatic
Max Tx Rate: 300Mbps
MAC Address: 90-F6-52-69-7C-B2
WDS Status: Disable

WAN

MAC Address: 90-F6-52-69-7C-B3
IP Address: 0.0.0.0
Subnet Mask: 0.0.0.0
Default Gateway: 0.0.0.0
DNS Server: 0.0.0.0, 0.0.0.0

Dynamic IP

Renew

Status Help

The Status page displays the Router's current status and configuration. All information is read-only.

LAN - The following parameters apply to the LAN port of the Router. You can configure them in the **Network > LAN** page:

- MAC Address - The physical address of the Router, as seen from the LAN.
- IP Address - The LAN IP address of the Router.
- Subnet Mask - The subnet mask associated with LAN IP address.

Wireless - These are the current settings or information for Wireless. You can configure them in the **Wireless > Wireless Settings** page:

- Wireless Radio - Indicates whether the wireless radio feature of the Router is enabled or disabled.
- Name (SSID) - The SSID of the Router.
- Channel - The current wireless channel in use.
- Mode - The current wireless mode which the Router works on.
- Channel Width - The bandwidth of the wireless channel.
- Max Tx Rate - The maximum Tx rate.
- MAC Address - The physical address of the Router, as seen from the WLAN.
- WDS Status - The status of WDS connection, the WDS connection is down; Scan: Try to find the AP; Auth: Try to authenticate; ASSOC: Try to associate; Run: Associated successfully.

WAN - The following parameters apply to the WAN ports of the Router. You can configure them in the **Network > WAN** page:

- MAC Address - The physical address of the WAN port, as seen from the Internet.
- IP Address - The current WAN (Internet) IP Address. This field will be blank or 0.0.0.0 if the IP Address is assigned dynamically and there is no connection to Internet.
- Subnet Mask - The subnet mask associated with the WAN IP Address.
- Default Gateway - The Gateway currently used by the Router is shown here. When you use Dynamic IP as the connection Internet type, the Renew button will be displayed here. Click the Renew button to obtain new IP parameters dynamically from the ISP. And if you have got an IP address, Release button will be displayed here. Click the Release button to release the IP address the Router has obtained from the ISP.
- DNS Server - The DNS (Domain Name System) Server IP addresses currently used by the Router. Multiple DNS IP settings are common. Usually, the first available DNS Server is used.
- Online Time - The time that you online. When you use PPPoE as WAN connection type, the online time is displayed here. Click the Connect or Disconnect button to connect to or disconnect from Internet.

Secondary Connection - Besides PPPoE, if you use an extra connection type to connect to a local area network provided by ISP, then parameters of this secondary connection will be shown in this area.

Traffic Statistics - The Router's traffic statistics:

- Sent (Bytes) - Traffic that counted in bytes has been sent out from the WAN port.
- Sent (Packets) - Traffic that counted in packets has been sent out from WAN port.
- Received (Bytes) - Traffic that counted in bytes has been received from the WAN port.
- Received (Packets) - Traffic that counted in packets has been received from the WAN port.

System Up Time - The length of the time since the Router was last powered on or reset.

Click the Refresh button to get the latest status and settings of the Router.

In our case, we're going to create a wireless network for our mobile devices. So, we'll access the WAN and DHCP settings, where we enter the necessary IP addresses assigned to our group.

DHCP Settings

DHCP Server: ☐ Disable ☒ Enable

Start IP Address:

End IP Address:

Address Lease Time: minutes (1~2880 minutes, the default value is 120)

Default Gateway: (optional)

Default Domain: (optional)

Primary DNS: (optional)

Secondary DNS: (optional)

The change of DHCP config will not take effect until the Router reboots, please [click here](#) to reboot.

Save

WAN

WAN Connection Type:

IP Address:

Subnet Mask:

Default Gateway: (Optional)

MTU Size (in bytes): (The default is 1500, do not change unless necessary.)

Primary DNS: (Optional)

Secondary DNS: (Optional)

Save

Now, the router is connected to the network. Therefore, it's necessary to configure the wireless network security by entering the password and WPA-PSK encryption and their respective settings.

☐ **Disable security**

☐ **WEP**

Type:

WEP Key Format:

Key Selected	WEP Key (Password)	Key Type
Key 1: ☒		Disabled
Key 2: ☐		Disabled
Key 3: ☐		Disabled
Key 4: ☐		Disabled

☐ **WPA/WPA2 - Enterprise**

Version:

Encryption:

Radius Server IP:

Radius Port: (1-65535, 0 stands for default port 1812)

Radius Password:

Group Key Update Period: (in second, minimum is 30, 0 means no update)

☒ **WPA/WPA2 - Personal(Recommended)**

Version:

Encryption:

PSK Password:

(You can enter ASCII characters between 8 and 63 or Hexadecimal characters between 8 and 64.)

Group Key Update Period: Seconds (Keep it default if you are not sure, minimum is 30, 0 means no update)

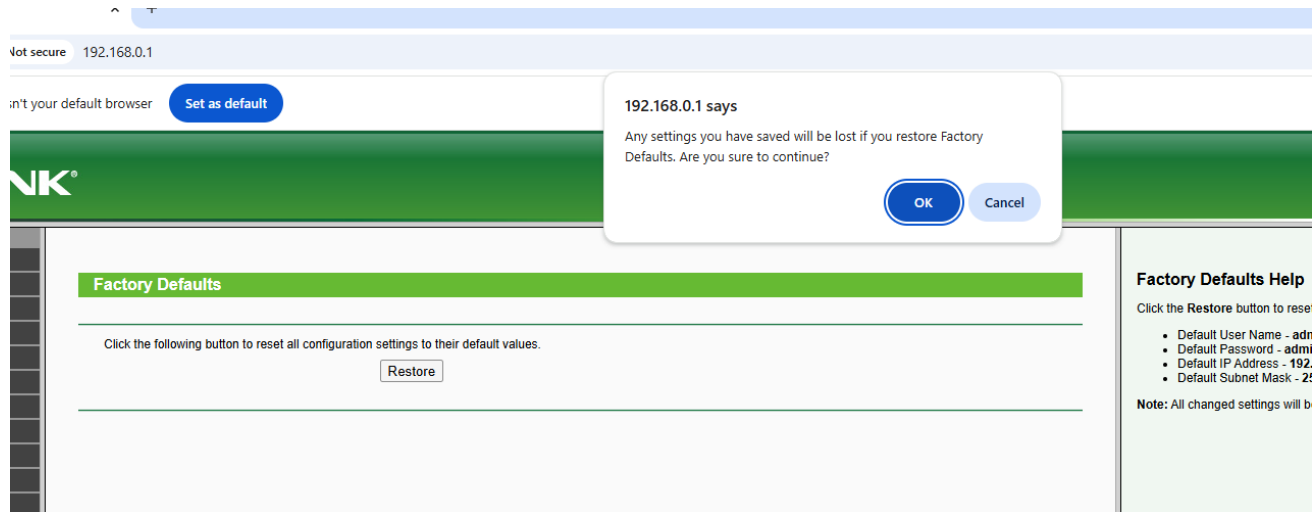
The change of wireless config will not take effect until the Router reboots, please [click here](#) to reboot.

Now, we restart the router and confirm our action to make the configuration changes effective. This process takes a while, but once it's finished, we should have a functioning network.

Restart

Restarting...

13%



Finally, we check the information about the deployed network and can see the created network on our mobile devices. This way, we can confirm that it's correctly finalized and ready to use.

Status

Firmware Version:3.13.9 Build 120201 Rel.54965n

Hardware Version:WR941N v2/v3 00000000

LAN

MAC Address:90-F6-52-89-7C-B2

IP Address:192.168.0.1

Subnet Mask:255.255.255.0

Wireless

Wireless Radio:Enable

Name (SSID):Lab8_Espinosa

Channel:11

Mode:11bgn mixed

Channel Width:Automatic

Max Tx Rate:300Mbps

MAC Address:90-F6-52-89-7C-B2

WDS Status:Disable

WAN

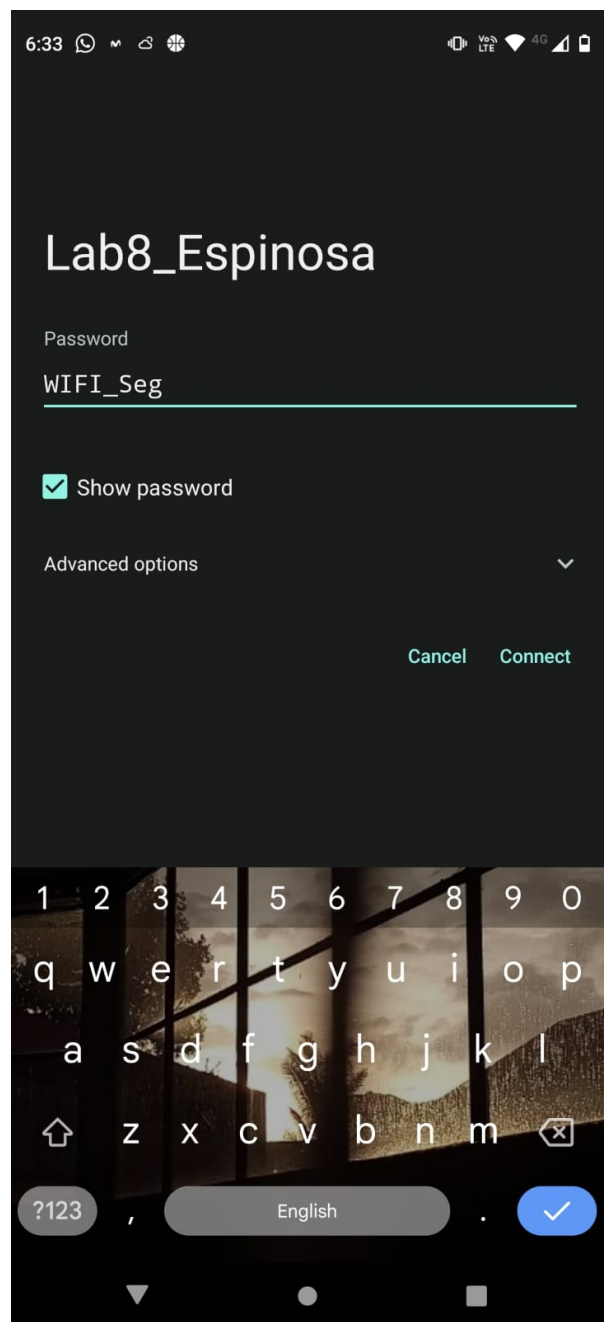
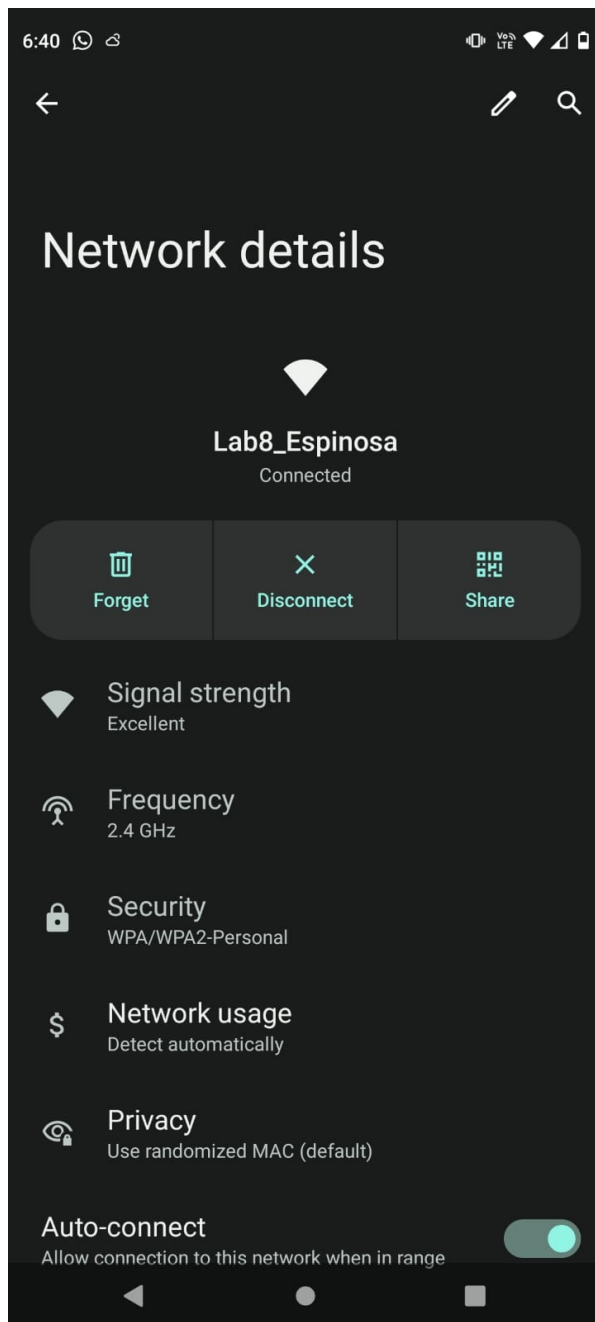
MAC Address:90-F6-52-89-7C-B3

IP Address:10.2.77.27Static IP

Subnet Mask:255.255.0.0

Default Gateway:10.2.65.1

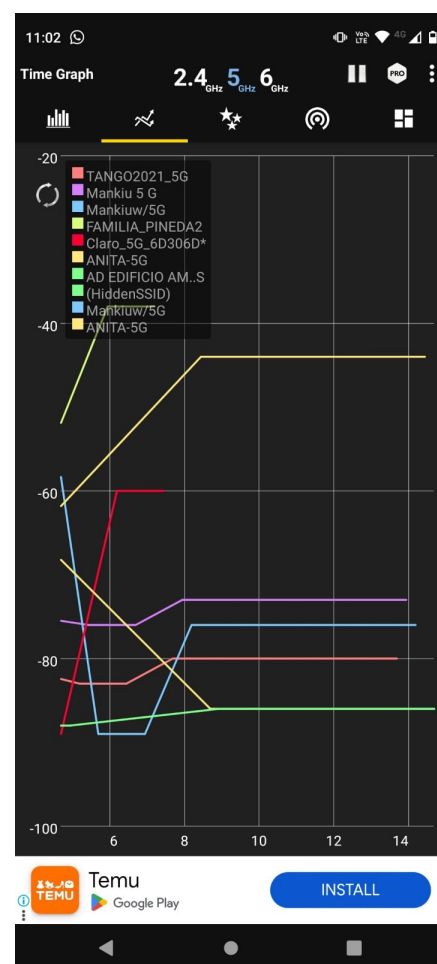
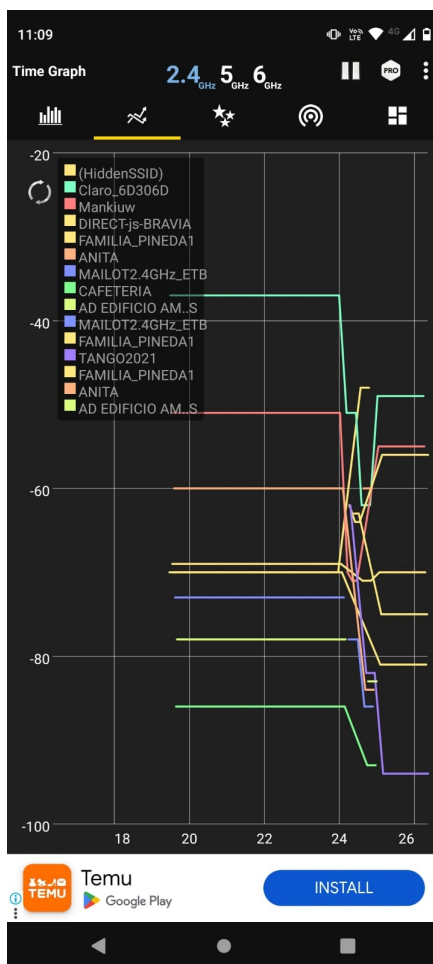
DNS Server:10.2.65.1 , 10.2.65.2

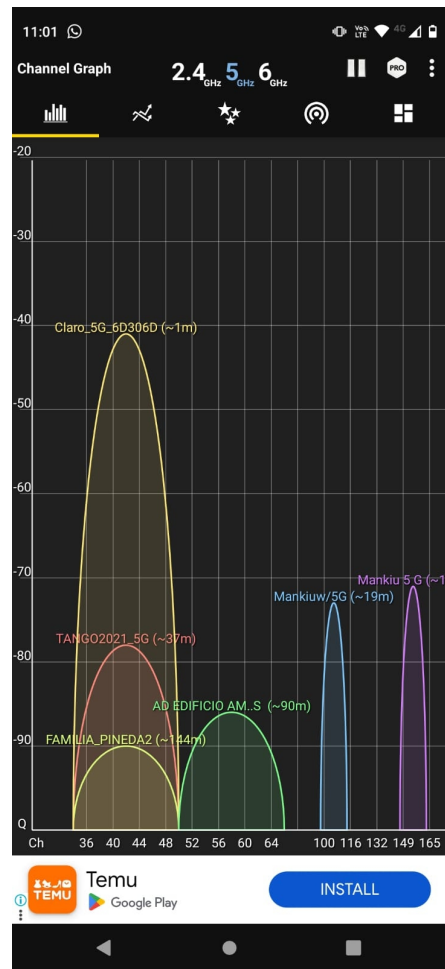
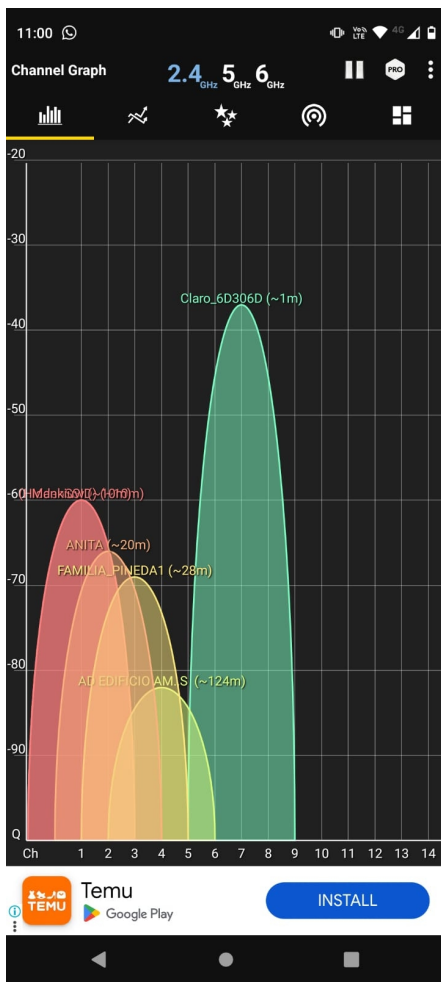


REVIEW WIFI NETWORKS NEAR YOUR HOME

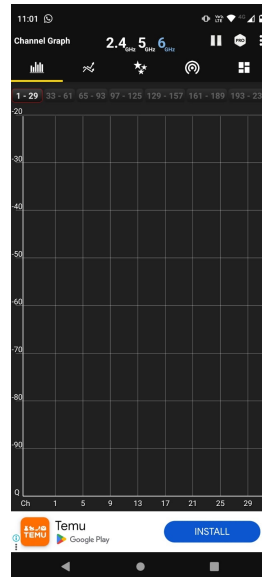
In this section of the lab, we are asked to use a wireless traffic monitoring app, such as WiFi Analyzer for Android, to explore the environment of nearby WiFi networks, including our own home network. The main objective is to identify available wireless networks in the area, collecting relevant information such as the network name, the frequency band in which they operate (2.4 GHz, 5.7 GHz, or 60 GHz), and the specific channels they use within those bands.

During the scan, a significant presence of networks operating in the 2.4 GHz band was found, although many of these had limited bandwidth and were subject to greater interference due to the common saturation of this band. On the other hand, networks operating in 5 GHz were the most predominant and prominent, as they offer better performance thanks to a greater number of available channels, less interference, and higher speeds in environments with good coverage.





Regarding the 60 GHz band, no active networks were detected. This is expected, as this band is associated with newer technologies such as WiGig (802.11ad/ay), which are not yet widely deployed in home environments due to their high hardware requirements and limited coverage, given that 60 GHz waves do not easily pass through walls.



This exercise allowed us not only to identify the characteristics of the wireless environment, but also to understand how technologies vary depending on the operating band, coverage, speed, and level of adoption—critical aspects for the analysis and design of efficient Wi-Fi networks.



BASE SOFTWARE INSTALLATION

In this section, a dynamic web service is deployed using Apache and PHP on a cloud infrastructure. An application is developed to calculate final grades and store the data in a relational database, such as PostgreSQL. This practice allows you to apply application layer concepts and strengthen the interaction between web services and databases.

DYNAMIC WEB SERVICE

In this part of the lab, we focus on building and deploying a simple web application using an Ubuntu virtual machine hosted on AWS EC2. The main idea is to create a functional grade calculator that allows students to input their name and the grades for each third of the semester. The app then calculates the final grade based on a 30%, 30%, and 40% distribution, providing an immediate and clear result.

We begin by launching and configuring the EC2 instance. This includes updating the system, opening the necessary ports for HTTP traffic, and setting up the Apache web server to host our application. Once the server environment is ready, we install PHP and other required packages so the server can dynamically interpret and run our PHP-based webpage.

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Buscar más AMI

Inclusión de AMI de AWS, Marketplace y la comunidad

Imágenes de máquina de Amazon (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-084568db4383264d4 (64 bits (x86)) / ami-0c4e709339fa8521a (64 bits (Arm))
Virtualización: hvm Activado para ENA: true Tipo de dispositivo raíz: ebs

Apto para la capa gratuita

Descripción

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Arquitectura
64 bits (x86)

ID de AMI
ami-084568db4383264d4

Fecha de publicación
2025-03-05

Nombre de usuario
ubuntu

Proveedor verificado

EC2

>

Instancias

>

i-06244d69c0af86ec5

>

Conectarse a la instancia

Conexión de la instancia EC2

Administrador de sesiones

Cliente SSH

Consola de serie de EC2

ID de la instancia

i-06244d69c0af86ec5

(PromedioDePasar)

Tipo de conexión

☒ Conectarse mediante la Conexión de la instancia EC2
Conéctese mediante el cliente basado en navegador de EC2 Instance Connect, con una dirección IPv4 o IPv6 pública.

☐ Conectarse mediante punto de conexión de EC2 Instance Connect
Conéctese mediante el cliente basado en navegador de EC2 Instance Connect, con una dirección IPv4 privada y un punto de conexión de VPC.

Dirección IPv4 pública

34.224.223.100

Dirección IPv6

Nombre de usuario

Escriba el nombre de usuario definido en la AMI utilizada para lanzar la instancia. Si no definió un nombre de usuario personalizado, utilice el nombre de usuario predeterminado, ubuntu.

ubuntu

X

Nota:

En la mayoría de los casos, el nombre de usuario predeterminado, ubuntu, es correcto. Sin embargo, lea las instrucciones de uso de la AMI para comprobar si el propietario de la AMI ha cambiado el nombre de usuario predeterminado.

```

ubuntu@ip-172-31-29-252:~$ sudo apt systemctl start apache2
E: Invalid operation systemctl
ubuntu@ip-172-31-29-252:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-29-252:~$ sudo systemctl start apache2
ubuntu@ip-172-31-29-252:~$ sudo systemctl enable apache2
}}Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-29-252:~$

```

```

Verifying : libsodium-1.0.19-4.amzn2023.x86_64 10/25
Verifying : libxslt-1.1.43-1.amzn2023.0.1.x86_64 11/25
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 12/25
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 13/25
Verifying : mod_lua-2.4.62-1.amzn2023.x86_64 14/25
Verifying : nginx-f-filesystem-1:1.26.3-1.amzn2023.0.1.noarch 15/25
Verifying : php8.4-8.4.5-1.amzn2023.0.1.x86_64 16/25
Verifying : php8.4-cli-8.4.5-1.amzn2023.0.1.x86_64 17/25
Verifying : php8.4-common-8.4.5-1.amzn2023.0.1.x86_64 18/25
Verifying : php8.4-fpm-8.4.5-1.amzn2023.0.1.x86_64 19/25
Verifying : php8.4-mbstring-8.4.5-1.amzn2023.0.1.x86_64 20/25
Verifying : php8.4-opcache-8.4.5-1.amzn2023.0.1.x86_64 21/25
Verifying : php8.4-pdo-8.4.5-1.amzn2023.0.1.x86_64 22/25
Verifying : php8.4-process-8.4.5-1.amzn2023.0.1.x86_64 23/25
Verifying : php8.4-sodium-8.4.5-1.amzn2023.0.1.x86_64 24/25
Verifying : php8.4-xml-8.4.5-1.amzn2023.0.1.x86_64 25/25

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.1.x86_64      apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.62-1.amzn2023.x86_64      httpd-core-2.4.62-1.amzn2023.x86_64
httpd-f-filesystem-2.4.62-1.amzn2023.noarch      httpd-tools-2.4.62-1.amzn2023.x86_64      libbrotli-1.0.9-4.amzn2023.0.2.x86_64
libsodium-1.0.19-4.amzn2023.x86_64             libxslt-1.1.43-1.amzn2023.0.1.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-2.0.27-1.amzn2023.0.3.x86_64         mod_lua-2.4.62-1.amzn2023.x86_64         nginx-f-filesystem-1:1.26.3-1.amzn2023.0.1.noarch
php8.4-8.4.5-1.amzn2023.0.1.x86_64             php8.4-cli-8.4.5-1.amzn2023.0.1.x86_64    php8.4-common-8.4.5-1.amzn2023.0.1.x86_64
php8.4-fpm-8.4.5-1.amzn2023.0.1.x86_64         php8.4-mbstring-8.4.5-1.amzn2023.0.1.x86_64  php8.4-opcache-8.4.5-1.amzn2023.0.1.x86_64
php8.4-pdo-8.4.5-1.amzn2023.0.1.x86_64         php8.4-process-8.4.5-1.amzn2023.0.1.x86_64  php8.4-sodium-8.4.5-1.amzn2023.0.1.x86_64
php8.4-xml-8.4.5-1.amzn2023.0.1.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-19-63 ~]$

```

Next, we develop the application itself. The interface is straightforward: students can enter their name and three grades into a simple form. When the form is submitted, the PHP code processes the input, performs the weighted average calculation, and displays the final result directly on the webpage. This interaction makes the app both functional and easy to use.

```
API Gateway
GNU nano 8.3 /var/www/html/index.php Modified
<?php
$final = null;
$message = '';

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $nombre = $_POST['nombre'];
    $notal = floatval($_POST['notal1']);
    $nota2 = floatval($_POST['nota21']);
    $nota3 = floatval($_POST['nota31']);

    $final = round(($notal * 0.3) + ($nota2 * 0.3) + ($nota3 * 0.4), 2);

    // Guardar en base de datos
    try {
        $pdo = new PDO("pgsql:host=<HOST>;port=5432;dbname=<DBNAME>", "<USUARIO>", "<CONTRASENA>");
        $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
        $stmt = $pdo->prepare("INSERT INTO notas (nombre, notal, nota2, nota3, final) VALUES (?, ?, ?, ?, ?)");
        $stmt->execute([$nombre, $notal, $nota2, $nota3, $final]);
        $message = "Datos guardados correctamente.";
    } catch (PDOException $e) {
        $message = "Error al guardar: " . $e->getMessage();
    }
}

<!DOCTYPE html>
<html>

G Help      O Write Out  F Where Is   K Cut       T Execute   C Location  M-U Undo    M-A Set Mark M-] To Bracket
X Exit      R Read File   A Replace    U Paste     J Justify   / Go To Line M-E Redo    M-6 Copy     B Where Was
```

To store the submitted data, we integrate a relational database. We initially attempt to install PostgreSQL directly on the machine but face some complications. To work around this, we use Docker to containerize the PostgreSQL database, which simplifies management and improves compatibility. The Dockerized database connects smoothly with our application, allowing it to save and retrieve data effectively.

```
[ec2-user@ip-172-31-19-63 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service -> /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-19-63 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-19-63 ~]$ newgrp docker
[ec2-user@ip-172-31-19-63 ~]$ docker run -it --rm postgres \
    psql -h reco.ctv4gz1luto4.us-east-1.rds.amazonaws.com -U postgres -d postgres
Unable to find image 'postgres:latest' locally
latest: Pulling from library/postgres
254e724d7786: Pull complete
074b41e8190f: Pull complete
bb26461722d7: Pull complete
0ddf304e15c0: Pull complete
693761670bfc: Pull complete
fb0c607b2495: Pull complete
f438059a5a57: Pull complete
64d10e07b30c: Pull complete
877e6448d55b: Pull complete
e023e132da3e: Pull complete
3ee9d9321249: Pull complete
a308d0de7d38: Pull complete
4d7ebd47166d: Pull complete
4ca3b919b7ab: Pull complete
Digest: sha256:864831322bf2520e7d03d899b01b542de6de9ece6f e29c89f19dc5e1d5568ccf
Status: Downloaded newer image for postgres:latest
Password for user postgres: |
```

```
postgres=> CREATE DATABASE reco;
CREATE DATABASE
postgres=> \c reco;
psql (17.5 (Debian 17.5-1.pgdgl20+1), server 17.2)
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off, ALPN: postgresql)
You are now connected to database "reco" as user "postgres".
reco=> CREATE TABLE notas (
    id SERIAL PRIMARY KEY,
    nombre VARCHAR(100) NOT NULL,
    parcial1 NUMERIC(5,2),
    parcial2 NUMERIC(5,2),
    parcial3 NUMERIC(5,2),
    final NUMERIC(5,2)
);
CREATE TABLE
```

After putting everything together, we test the application thoroughly. It performs well—the calculation logic is accurate, the data is successfully stored in the database, and the overall system runs reliably on the cloud. This experience gives us valuable insight into full-stack development, cloud deployment, and the practical use of tools like Apache, PHP, PostgreSQL, and Docker in a real-world scenario.

Calculadora de Notas del Semestre

Nombre del estudiante:

Nota primer tercio (30%):

Nota segundo tercio (30%):

Nota tercer tercio (40%):

CONCLUSIONS

This lab allowed us to gain in-depth and practical knowledge of key network and application layer components, essential for the functioning of modern technological infrastructures. During the activities, we configured both wired and wireless networks, giving us a clearer understanding of how devices such as switches, routers, and access points operate, and how they interact at the data link layer to ensure effective data transmission.

One of the highlights was the configuration and analysis of Ethernet frames using Wireshark, which allowed us to study frame structure, MAC addresses, error control, and the process of frame learning and forwarding by switches. This gave us a detailed look at how switches handle network traffic at the data link layer, including the packet broadcasting process.

The implementation of VLANs was also critical, as it allowed us to segment the network into different broadcast domains, improving network security and performance. We effectively configured VLANs, assigning devices to different network segments, and set up trunking to allow traffic transmission between different switches and VLANs. This exercise reinforced our understanding of network segmentation concepts and the advantages of using VLANs in an enterprise infrastructure.

In the wireless networking segment, we configured a secure Wi-Fi network using WPA2-PSK with AES and learned how to manage mobile devices within this network. Analyzing nearby networks using applications such as Wi-Fi Analyzer also allowed us to explore channel management and security settings for wireless networks.

Furthermore, creating and configuring a dynamic web service, including integrating PHP with relational databases such as PostgreSQL, allowed us to better understand how web applications work and how they interact with database servers. This application layer component also involved implementing a shell script that executes and displays network commands, facilitating network administration and monitoring using tools such as ifconfig, netstat, vnstat, and route, among others.

In conclusion, this lab not only provided us with practical knowledge of network configuration and administration but also allowed us to understand the fundamental aspects of the interaction between the data link layer and the application layer, which is essential for designing, managing, and monitoring network infrastructures in the real world. This experience has given us a solid foundation for applying these concepts in professional situations, improving our ability to manage networks and services efficiently and securely.

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